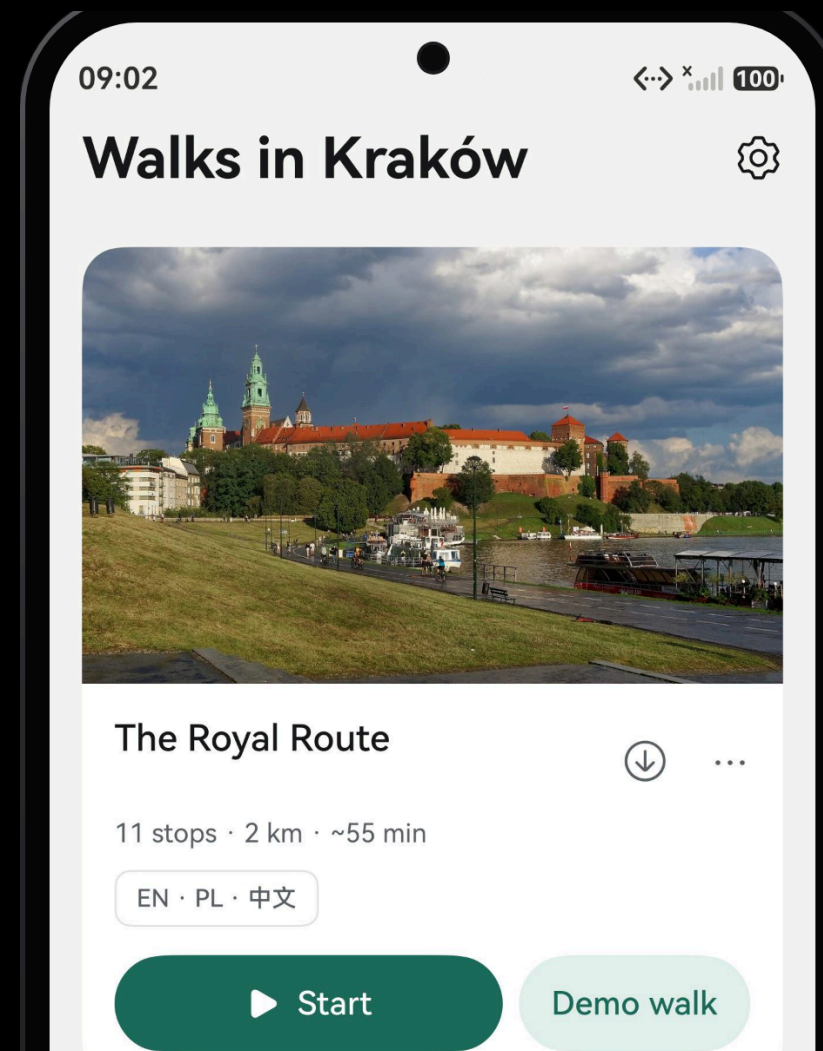




CityTour

Kraków's stories, told where they happened.



Every city is full of stories.

Most visitors walk right past them.

The problem

Today, you have to choose.

Guided tours

Great stories. But they cost money, start at fixed times and move at the group's pace.

Apps and guidebooks

Cheap. But your eyes end up on a screen instead of the city.

Walking alone

Free and flexible. But the history around you stays invisible.

Introducing CityTour

Your guide fits in your pocket.

Put your headphones in and keep your eyes on the city.
CityTour plans your route, follows you through the city and
tells each place's story as you reach it.



Demo

See it running.

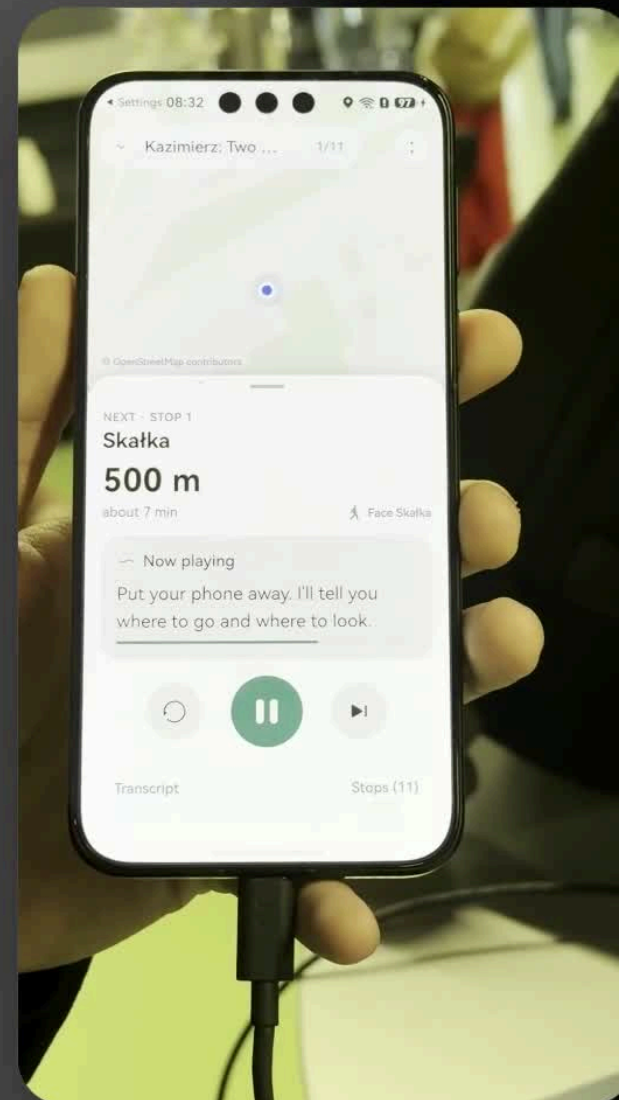
Real Huawei phone · live GPS · Kraków

**Tap Start.
Then put it away.**



GEORGE · ON THE PHONE

“Put your phone away. I’ll tell you where to go and where to look.”



Real Huawei phone

Live GPS at the venue in Kraków

Emulator walk

Arrival story, turn-by-turn, screen off

Three languages

English, Polski, Chinese, mid-story



66-second video
Scan or click to play

Real app, not mock-ups. The Demo walk is labelled SIMULATED.

Choose a walk

Three walks through Kraków.



The Royal Route

11 stops · 2 km · ~55 min



Scholars and Saints

10 stops · ~52 min

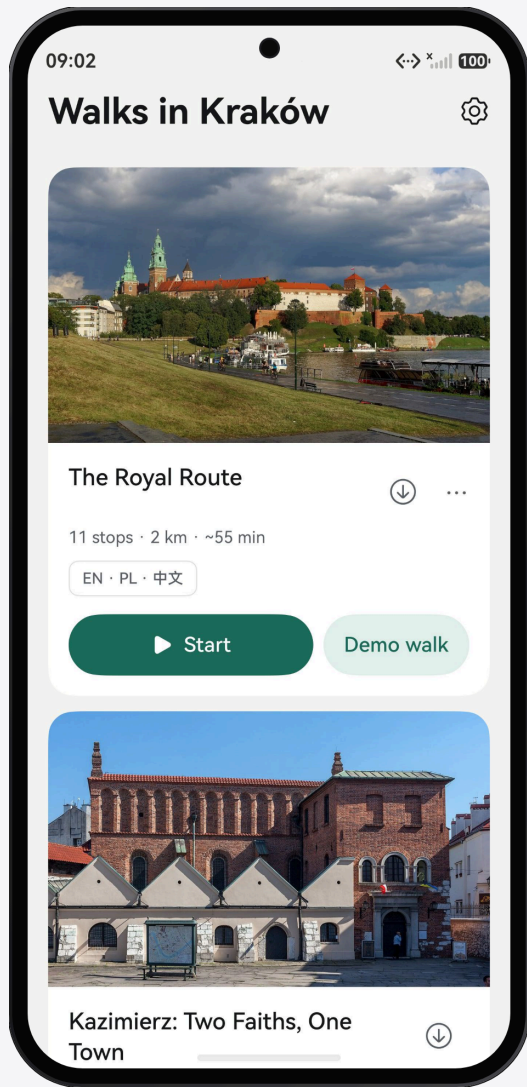


Kazimierz: Two Faiths, One Town

11 stops · ~57 min

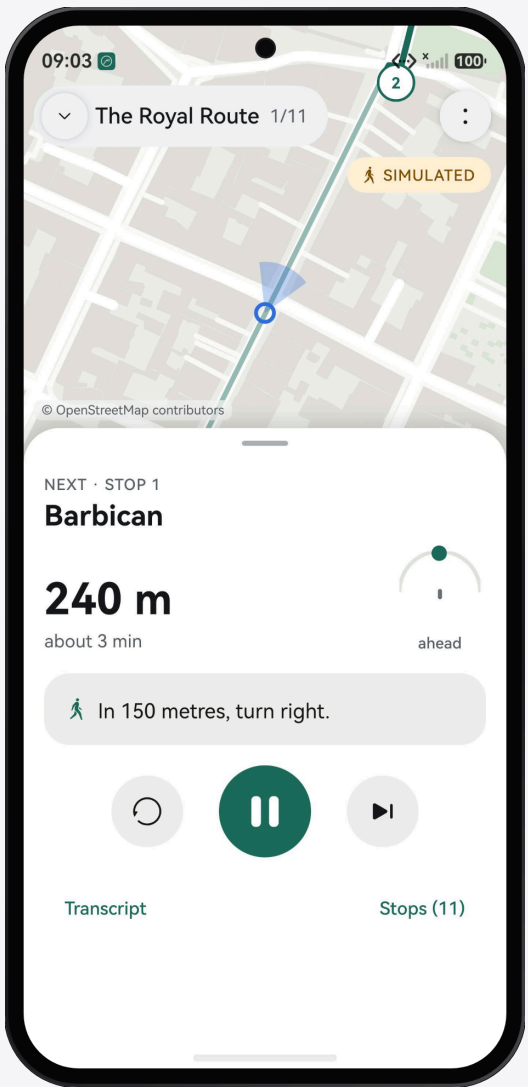
How it works

Four steps. One pocket.



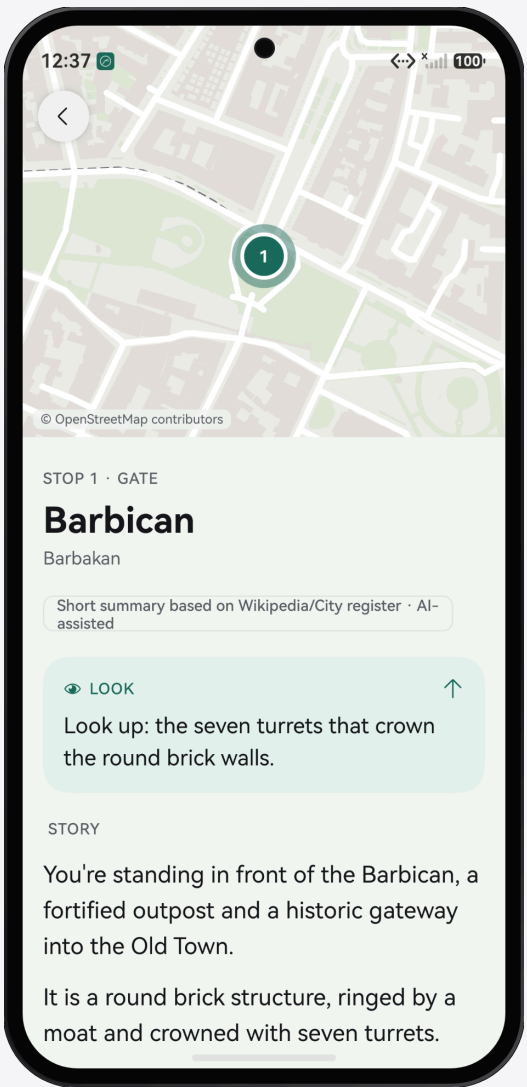
Tap Start

The best order, planned from where you stand.



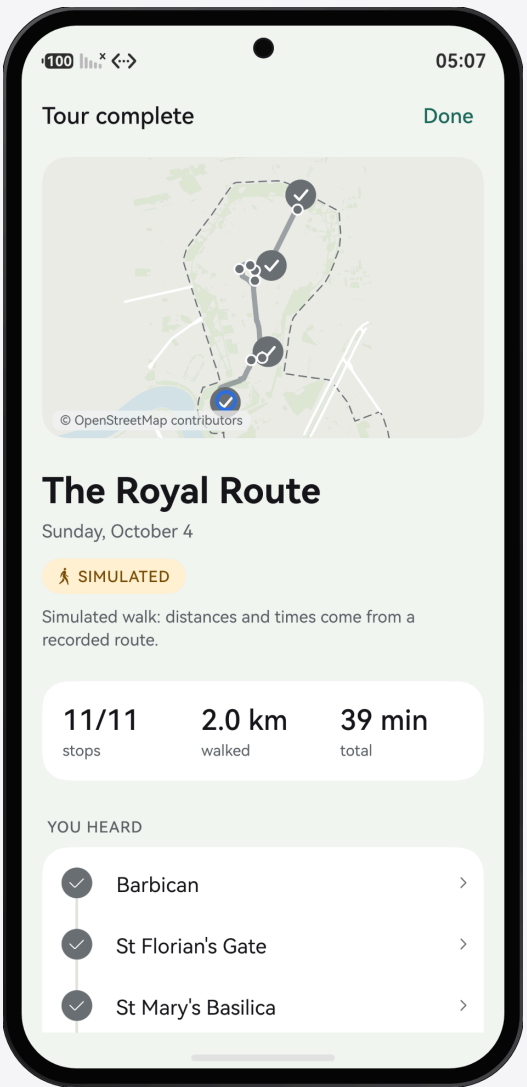
Eyes on the city

The guide talks you there, turn by turn.



Arrive and listen

Where to look, then the story.



Finish

Every story you heard, in one place.

Our challenge area

Spatial Experiences.

Where you are, and which way you walk, decide what the guide says, when it says it, and where you should look.

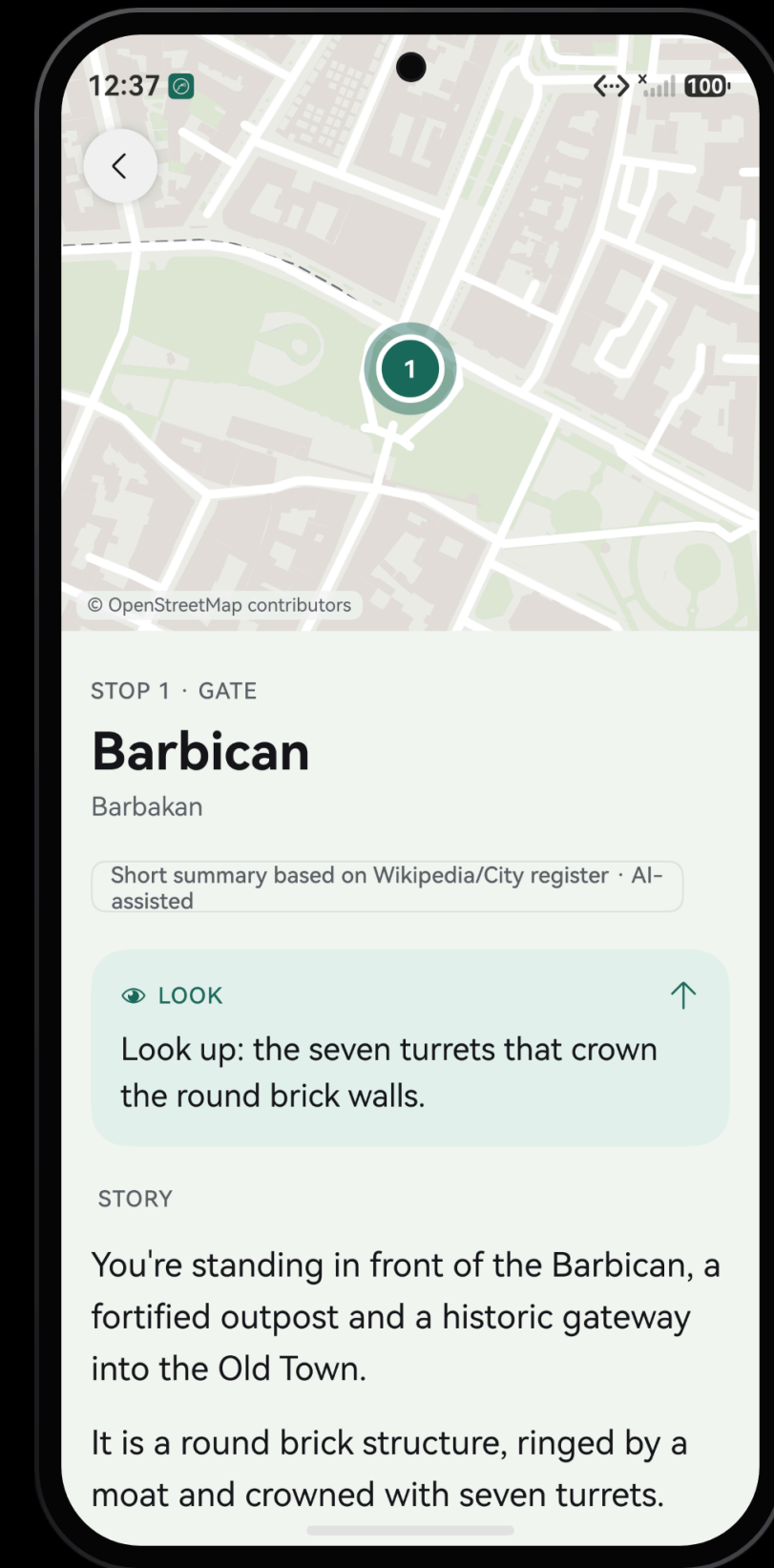
Also Human-Centric Technology: a cultural experience in three languages that keeps your eyes on the city, not the screen.

Spatial

Your guide points it out.

"Look up: the seven turrets that crown the round brick walls."

CityTour knows which way you face from the way you walk, so the phone can stay in your pocket.



Context-aware

It listens to how you walk.

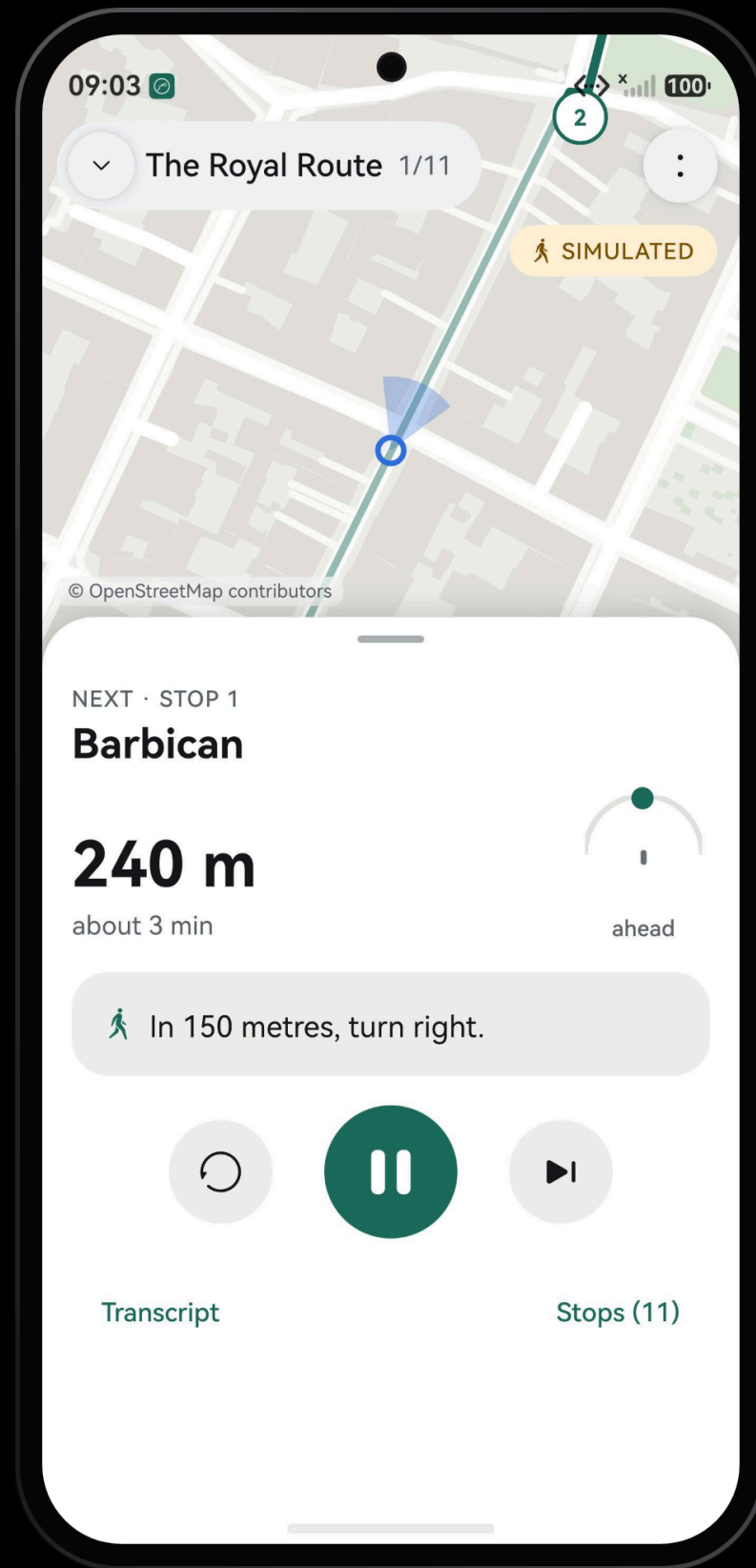
Stop.

You hear the full story.

Walk past.

You get the short version.

Your walking speed decides. No buttons, no screen.



Turn-by-turn

“In 150 metres,
turn right.”

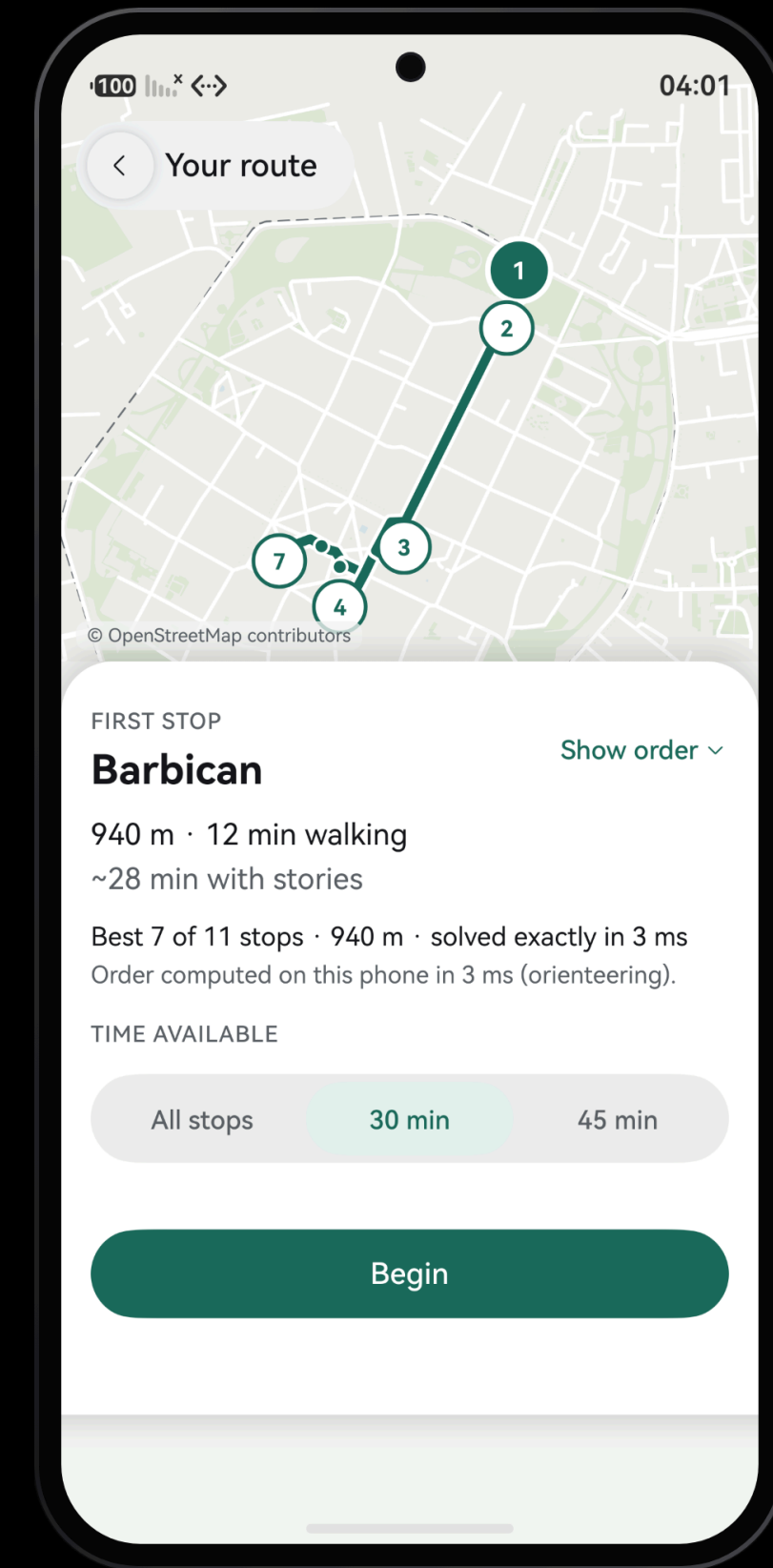
Spoken directions between stops. Wander off the route, and CityTour notices and re-plans from where you are.

Short on time?

30 min

It picks the best 7 of 11 stops.

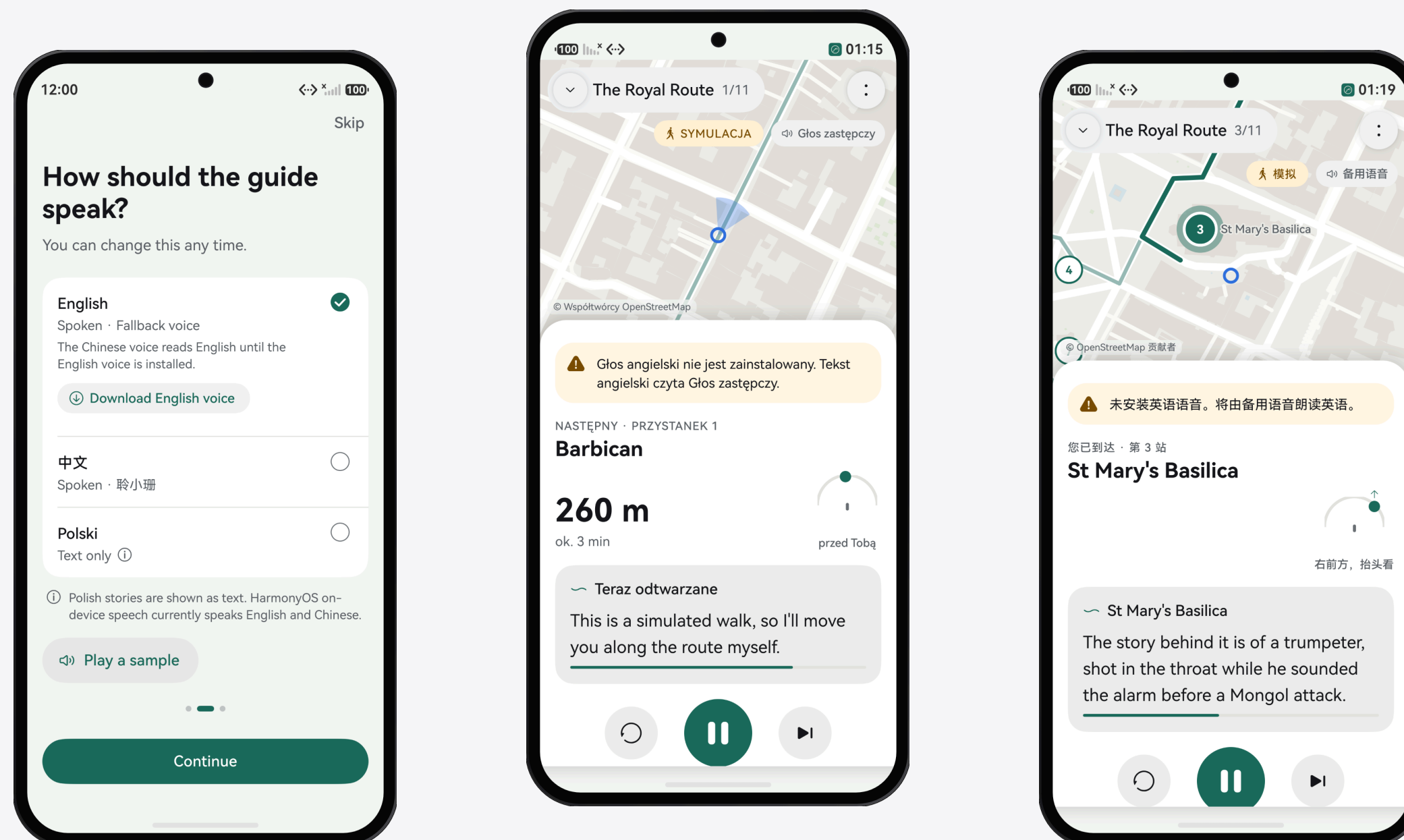
Exact route planning over real walking times, solved on the phone in milliseconds.



Studio voice

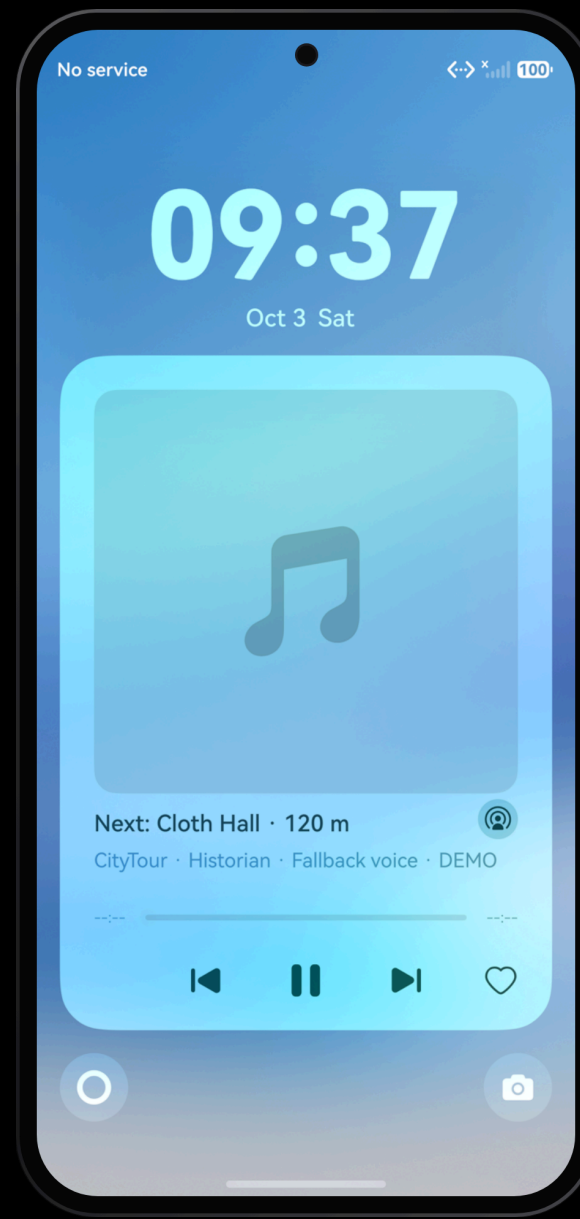
A real voice. In three languages.

3,524 studio-voice clips across the three walks, in English, Polski and 中文. Switch language mid-story without missing a sentence.

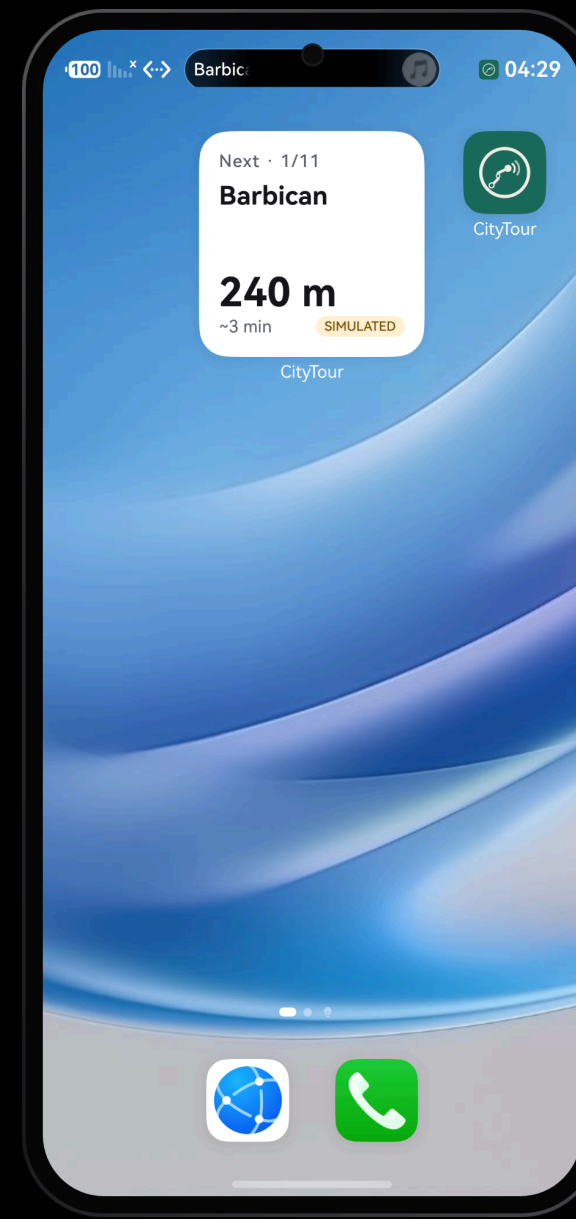


Always a glance away

Lock screen. Home screen. Wrist.



Lock screen



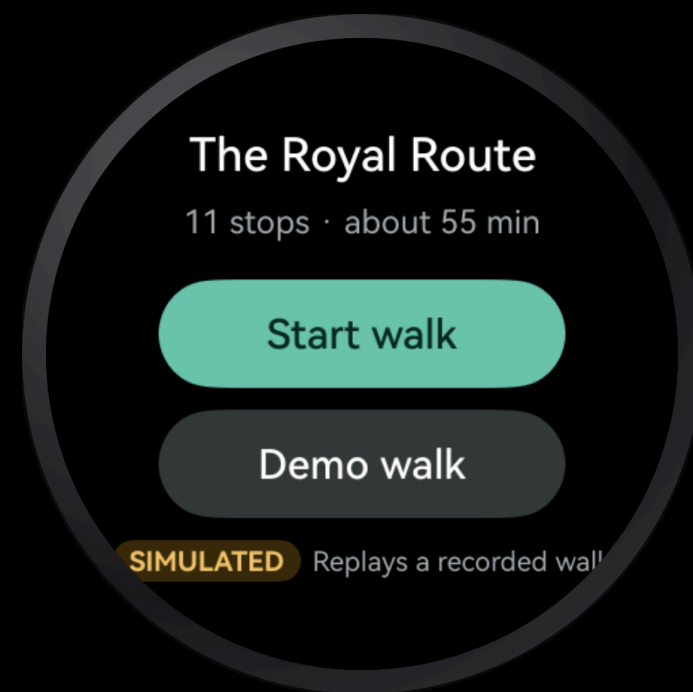
Home-screen widget



Huawei watch

Huawei watch

The whole tour, on your wrist.



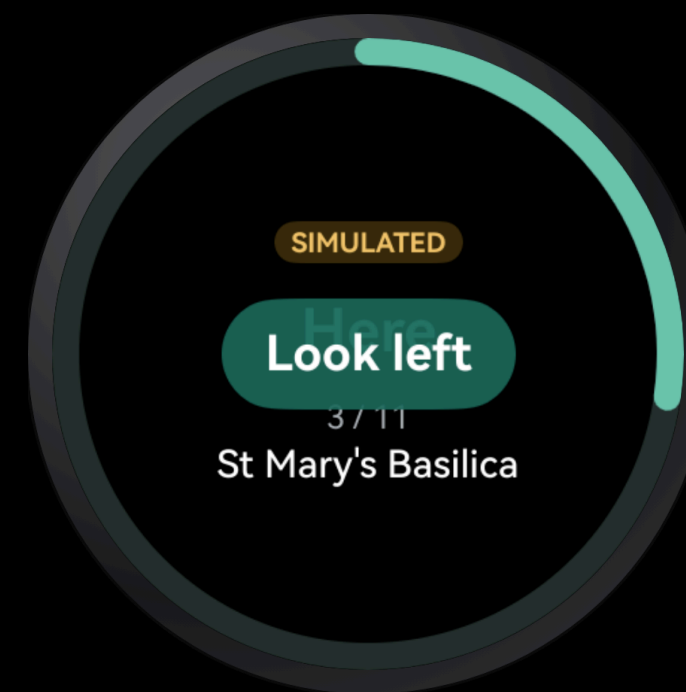
Start

The Royal Route, ready on the watch.



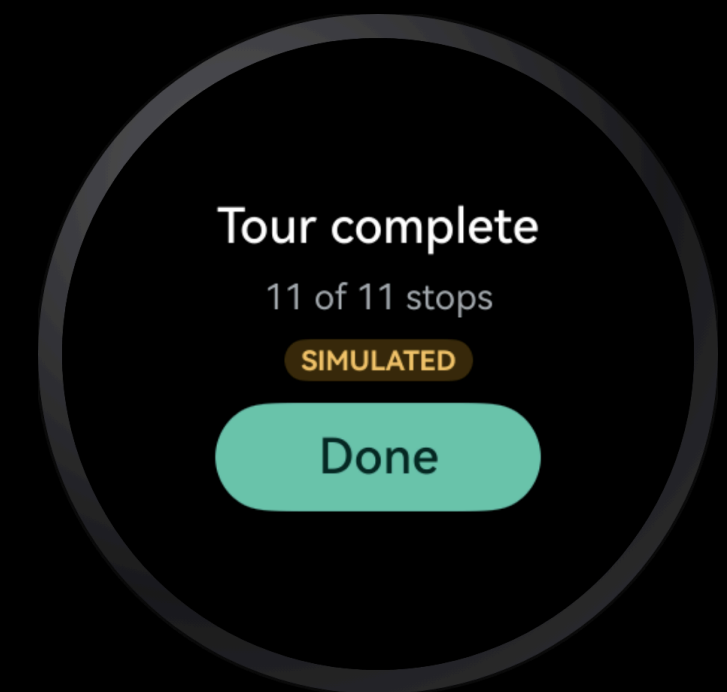
Follow the arrow

Distance and direction to the next stop.



Feel where to look

2 taps: look left. 3 taps: look right.



Done

11 of 11 stops.

Download once. Walk without signal.

Start a walk instantly, or keep it offline: maps, stories and studio voice stay on the phone. Every download is signed and verified. No account, and your location never leaves your phone.

Platform

Built deep into HarmonyOS.

Location Kit

Position and walking direction, phone and watch.

Background Tasks

The tour keeps running with the screen locked.

AVSession Kit

Lock-screen and headset controls.

Media Kit

Plays the studio-voice clips.

Core Speech Kit

On-device voice when there's no clip.

Form Kit

The live Next stop home-screen widget.

Notification Kit

The next stop with the screen off.

Sensor Service Kit

Arrival and look-left / look-right haptics.

Network Kit

Streams and downloads walks, with deadlines.

Crypto Architecture

Ed25519 signatures and SHA-256 checks.

Core File Kit

Safe, atomic installs of each walk.

Localization Kit

English, Polski and 中文 interface.

Architecture

Two apps. One engine.

Build time

Data pipeline

OpenStreetMap, Wikidata, Wikipedia and OSRM walking routes, plus ElevenLabs studio voice, packed into city and walk packs.



Live

Course server

Express and TypeScript. Signed catalog, city packs, walks and voice clips. Renders any missing line once.



Phone app

Screens, voice, lock screen, widget

Watch app

Arrow, glance, haptic cues

Shared core

Tour engine, route planner, geo and content. No platform imports, fully unit-tested.

Open platform

Open data. Open core.

Open data

Maps, places and stories come from OpenStreetMap, Wikidata and Wikipedia, credited in the app.

A portable core

The tour engine has no platform imports, and our tests keep it that way. It's built to move to OpenHarmony and Oniro.

Walks are data

A new walk or city is a pack on the server, not new code. No gatekeeper decides what gets told.

HarmonyOS-only kits, such as Core Speech Kit, sit in adapters outside the core.

Engineering

Built to actually work.

438

unit tests, all passing

6

full replays of a recorded walk:
every stop in order, no sentence
cut off

17

failure cases, each with a clear
on-screen state instead of a
crash

2 apps

phone and watch, one shared
tour engine

1 server

live, signing every walk it delivers

400+

commits since Saturday, all public

Built at HackYeah

From an empty template, in 14 hours.

Sat 13:07

First commit: the empty
DevEco template

Sat 14:10

Architecture and platform
tests on the emulator

Sat 17:18

Kraków data pipeline

Sat 19:09

Studio voice and spoken
Polish

Sat 21:54

Course server

Sat 23:54

Three walks through Kraków

Sun 02:16

Huawei watch app

Sun 03:05

v1.0 release, with the .hap

Not ours: the DevEco Empty Ability scaffold and the challenge's starter files. Everything else was built here. Times from the public git history.

How we built it

Nothing hidden.

What we simulate

Location. The emulator's GPS is one fixed point, so a Demo walk replays a recorded route through the real tour engine.

Watch taps. The emulator has no vibration motor, so each cue also flashes on screen.

Every simulated screen, on phone and watch, is labelled **SIMULATED**.

Why the watch isn't linked

No emulator can show it. Sync uses Huawei's Wear Engine on real devices paired in Huawei Health, so it couldn't be checked.

It needs approval through AppGallery Connect, longer than a weekend.

Standalone works anyway: own GPS, own stops, no dropped connection.

How we worked

AI agents, in the open. Built with Claude Code; every task is logged in AI_WORKFLOW.md and 400+ public commits.

One rule: claim only what runs as described, and label anything mocked.

AI-drafted stories, checked against cited sources; not yet reviewed by a historian or native speakers.

Beyond the emulator

On a real device, it walks with you.

	On the emulator	On a real device
Location	The Demo walk replays a recorded route, labelled SIMULATED.	Live GPS from Location Kit, into the same tour engine. Switch it in Settings.
Watch cues	Each cue flashes on screen: the emulator has no motor.	You feel it: two taps for left, three for right.
Phone voice	Lines without a studio clip are read by the Chinese system voice.	The English system voice reads them, once it's installed.

Not yet run on real hardware. We'd love to try it on a mentor's device.

Reproducible

Try it yourself.

1

Get the code

`github.com/
Svetoslav47/citytour`

2

Build and launch

```
devecocli run  
--module entry  
--device "Pura 90"
```

3

Take a walk

Tap Demo walk on any walk. Or
install the .hap from the v1.0-
hackyeah release.

Public repository

Setup, build and launch in the README

Working .hap

Demo recording

Architecture docs

AI_WORKFLOW.md

What's next

One city today. Any city tomorrow.

More cities

The server already delivers city packs and walks. A new city is new data, not new code.

Phone + watch

A Wear Engine companion: the phone tells the story, the wrist taps. After Huawei's approval.

Tours by anyone

An open tour format that museums, schools and cities can publish themselves.



CityTour

Eyes on the city, not the screen.

github.com/Svetoslav47/citytour